Diskrete Mathematik HS2025 — Prof. Dennis HOFHEINZ

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Exercise sheet 5

This is the exercise sheet number 5. The difficulty of the questions and exercises are rated from very easy ($\star$) to hard ($\star\star\star\star$) . The graded exercise is Exercise 5.4 and your solution has to be uploaded on the Moodle page of the course **by 23/10/2025, 23:59**. The solution to this exercise must be your own work, you may not share your solutions with anyone else. See also the note on dishonest behavior on the Moodle page.

Exercise 5.1 Computing Representations of Relations ($\star$)

For the relation $\rho = \{(1,4),(2,1),(2,3),(4,2)\}$ on the set $\{1,2,3,4\}$, determine the relations ρ^3 and ρ^*. Describe ρ^3 using the set representation, and ρ^* using matrix representation (see lecture notes, Section 3.3.2).

Solution: We have $\rho^3 = \{(1,1),(1,3),(2,2),(4,4)\}$ and

$$M^{\rho^*} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Exercise 5.2 Operations on Relations ($\star\star$)

Let us consider the relations $<, |$ and $\equiv_2$ on the set of positive natural numbers $\mathbb{N} \setminus \{0\}$. For each of the following relations on $\mathbb{N} \setminus \{0\}$, decide whether it is reflexive, symmetric or transitive. Justify your answers.

1. $< \circ |$
2. $| \cup \equiv_2$
3. $| \cup \widehat{|}$

Solution:

	Relation	reflexive	symmetric	transitive
1.	$< \circ \mid$	✗	✗	✓
2.	$\mid \cup \equiv_2$	✓	✗	✗
3.	$\mid \cup \widehat{\mid}$	✓	✓	✗

1. Two numbers (a, b) are in the relation whenever there exists an x such that $a < x$ and $x \mid b$. This relation is not reflexive, since $(1, 1) \notin < \circ \mid$. Moreover, it is not symmetric, because $(1, 2) \in < \circ \mid$, but $(2, 1) \notin < \circ \mid$. This relation is transitive. For any (a, b, c), assume that there exist some x and y, such that $a < x$, $x \mid b$, $b < y$ and $y \mid c$. From $x \mid b$ it follows that $x \leq b$, hence, $a < x \leq b < y$. Therefore, $a < y$ and $y \mid c$.

2. Two numbers (a, b) are in the relation whenever $a \mid b$ or $a \equiv_2 b$. This relation is reflexive, since for any a, we have $a \equiv_2 a$ (alternatively, one could use the fact that $a \mid a$). It is, however, not symmetric, because $(1, 2) \in \ \mid \cup \equiv_2$, but $(2, 1) \notin \ \mid \cup \equiv_2$. It is also not transitive, since $(3, 1) \in \ \mid \cup \equiv_2$ and $(1, 2) \in \mid \cup \equiv_2$, but $(3, 2) \notin \ \mid \cup \equiv_2$.

3. Two numbers (a, b) are in the relation whenever $a \mid b$ or $b \mid a$. This relation is reflexive, since for any a, we have $a \mid a$. It is also symmetric, because for any (a, b), we trivially have $a \mid b$ or $b \mid a$ if and only if $b \mid a$ or $a \mid b$. The relation is, however, not transitive, since $(3, 1) \in \mid \cup \widehat{\mid}$ and $(1, 2) \in \mid \cup \widehat{\mid}$ but $(3, 2) \notin \mid \cup \widehat{\mid}$.

Exercise 5.3 An Equivalence Relation ($\star\ \star$)

The relation $\sim$ on $\mathbb{R}^2 \setminus \{(0, 0)\}$ is defined as follows:

$$(x_1, y_1) \sim (x_2, y_2) \quad \overset{\text{def}}{\iff} \quad \exists \lambda > 0\ (x_1, y_1) = (\lambda x_2, \lambda y_2)$$

1. Prove that $\sim$ is an equivalence relation.

Solution: We prove that $\sim$ satisfies all properties of an equivalence relation.

Reflexivity: For any point $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$, we have $(x, y) \sim (x, y)$, because one can choose $\lambda = 1$ in the definition of $\sim$.

Symmetry: Let $x_1, y_1, x_2, y_2 \in \mathbb{R} \setminus \{0\}$ and assume that $(x_1, y_1) \sim (x_2, y_2)$. It follows that $x_1 = \lambda x_2$ and $y_1 = \lambda y_2$ for some $\lambda > 0$. Hence, $x_2 = \frac{1}{\lambda} x_1$ and $y_2 = \frac{1}{\lambda} y_1$, where $\frac{1}{\lambda} > 0$. Therefore, $(x_2, y_2) \sim (x_1, y_1)$.

Transitivity: Let $x_1, y_1, x_2, y_2, x_3, y_3 \in \mathbb{R} \setminus \{0\}$ and assume that $(x_1, y_1) \sim (x_2, y_2)$ and $(x_2, y_2) \sim (x_3, y_3)$. This means that $(x_1, y_1) = (\lambda_1 x_2, \lambda_1 y_2)$ and $(x_2, y_2) = (\lambda_2 x_3, \lambda_2 y_3)$ for some $\lambda_1, \lambda_2 > 0$. It follows that $(x_1, y_1) = (\lambda x_3, \lambda y_3)$, where $\lambda > 0$ is defined as $\lambda_1 \lambda_2$. Hence, $(x_1, y_1) \sim (x_3, y_3)$.

2. Describe geometrically the equivalence classes $[(x, y)]$.

Solution: An equivalence class $[(x, y)]_\sim$ contains all points on the ray through the origin $(0, 0)$ and the point (x, y) (excluding the origin). Note that no equivalence class can contain the origin $(0, 0)$ ($\sim$ is only defined on $\mathbb{R}^2 \setminus \{(0, 0)\}$).

Exercise 5.4 Properties of Relations ($\star$) — GRADED *(8 points)*
Please upload your solution by 23/10/2025

Prove or disprove the following claims:

1. Let ρ be a relation on a set A. Then, the relation $\gamma \stackrel{\text{def}}{=} \rho \circ \widehat{\rho}$ is symmetric.

Solution: The claim is true, as the following holds for all $a \in A$ and $b \in A$:

$$\begin{aligned}
a\ \gamma\ b &\stackrel{\cdot}{\Longleftrightarrow} \exists c\ (a \rho c\ \wedge\ c \widehat{\rho} b) && \text{(definition of } \circ\text{)} \\
&\stackrel{\cdot}{\Longleftrightarrow} \exists c\ (a \rho c\ \wedge\ b \rho c) && \text{(definition of relation inverse)} \\
&\stackrel{\cdot}{\Longleftrightarrow} \exists c\ (b \rho c\ \wedge\ a \rho c) && \text{(commutativity of } \wedge\text{)} \\
&\stackrel{\cdot}{\Longleftrightarrow} \exists c\ (b \rho c\ \wedge\ c \widehat{\rho} a) && \text{(definition of relation inverse)} \\
&\stackrel{\cdot}{\Longleftrightarrow} b\ \gamma\ a && \text{(definition of } \circ\text{)}
\end{aligned}$$

2. Let A be a set and let σ be a symmetric relation on A and π be an antisymmetric relation on A. Then, the relation $\gamma \stackrel{\text{def}}{=} \sigma \circ \widehat{\pi}$ is symmetric.

Solution: The claim is false, which can be seen from the following counterexample: Let the set be $A \stackrel{\text{def}}{=} \mathbb{Z}$, let σ be the relation "$=$", and let π be the relation "$\geq$". Clearly σ is symmetric, and from the lecture notes (Example 3.28) we also know that π is antisymmetric. We will use the fact that $\widehat{\geq}$ is the same relation as $\leq$ (see Example 3.21 in the lecture notes). It holds that:

$$\begin{aligned}
1 = 1 \wedge 1 \leq 2 &\stackrel{\cdot}{\Longrightarrow} 1 = 1 \wedge 1 \mathrel{\widehat{\geq}} 2 \\
&\stackrel{\cdot}{\Longrightarrow} 1\ \gamma\ 2
\end{aligned}$$

However, $2\ \gamma\ 1$ is not true, because otherwise $\exists x\ (2 = x\ \wedge\ x \leq 1)$ would hold, which implies $2 \leq 1$, a contradiction. This shows that γ is not symmetric.

3. The intersection of two equivalence relations on the same set is an equivalence relation.

Solution: The claim is true. Let σ and ρ be two equivalence relations on a set A and let $\gamma \stackrel{\text{def}}{=} \sigma \cap \rho$. We will show that γ is reflexive, symmetric and transitive.

Reflexivity. The following holds for every $a \in A$: By reflexivity of σ and ρ, we have $a\ \sigma\ a$ and $a\ \rho\ a$. By definition of $\cap$, this implies $a\ \gamma\ a$.

Symmetry. The following holds for all $a \in A$ and $b \in A$:

$$\begin{aligned} a\ \gamma\ b &\stackrel{\cdot}{\Longleftrightarrow} a\ \sigma\ b\ \wedge\ a\ \rho\ b && \text{(definition of } \cap) \\ &\stackrel{\cdot}{\Longleftrightarrow} b\ \sigma\ a\ \wedge\ b\ \rho\ a && \text{(symmetry of } \sigma \text{ and } \rho) \\ &\stackrel{\cdot}{\Longleftrightarrow} b\ \gamma\ a && \text{(definition of } \cap) \end{aligned}$$

Transitivity. The following holds for all $a \in A$, $b \in A$, and $c \in A$:

$$\begin{aligned} a\ \gamma\ b \wedge b\ \gamma\ c &\stackrel{\cdot}{\Longrightarrow} (a\ \sigma\ b \wedge a\ \rho\ b)\ \wedge\ (b\ \sigma\ c \wedge b\ \rho\ c) && \text{(definition of } \cap) \\ &\stackrel{\cdot}{\Longrightarrow} (a\ \sigma\ b \wedge b\ \sigma\ c)\ \wedge\ (a\ \rho\ b \wedge b\ \rho\ c) && \text{(commutativity and associativity of } \wedge) \\ &\stackrel{\cdot}{\Longrightarrow} a\ \sigma\ c\ \wedge\ a\ \rho\ c && \text{(transitivity of } \sigma \text{ and } \rho) \\ &\stackrel{\cdot}{\Longrightarrow} a\ \gamma\ c && \text{(definition of } \cap) \end{aligned}$$

Exercise 5.5 Lifting an Operation to Equivalence Classes ($\star\ \star$)

In the lecture (and in Section 3.4.3 of the lecture notes), we have considered the following equivalence relation $\sim$ on the set $A \stackrel{\text{def}}{=} \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$:

$$(a, b) \sim (c, d) \iff ad = bc.$$

We then defined the rational numbers $\mathbb{Q} \stackrel{\text{def}}{=} A/\!\sim$, capturing the fact that each rational number has different **representations** as fractions. The goal of this exercise is to understand how to define an operation on equivalence classes (e.g., the sum of two rational numbers) by lifting an operation defined at the level of the **elements** of the equivalence classes (e.g., the fractional representations A of the rationals). Consider an equivalence relation θ on some set B and a function $f : B^2 \to B$. We want to lift f to the set of equivalence classes B/θ, i.e., we want to define a function $F : (B/\theta)^2 \to (B/\theta)$ canonically in terms of f. For this to be meaningful, f has to be θ**-consistent**. That is, if f is applied to a pair $(b_1, b_2) \in B^2$, the equivalence class $[f(b_1, b_2)]_\theta$ may only depend on the **equivalence classes** $[b_1]_\theta$ and $[b_2]_\theta$ (irrespective of which concrete elements both b_1 and b_2 are within their equivalence classes).

1. Define the sum of two fractional representations of rational numbers as a function $\mathsf{sum} : A^2 \to A$ (for $A = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ as defined above), using standard addition and multiplication in the integers.

Solution: We define the function $\mathsf{sum} : A^2 \to A$ by

$$\mathsf{sum}((a,b),(c,d)) \stackrel{\text{def}}{=} (ad+bc, bd).$$

Observe that $bd \neq 0$ since $b \neq 0$ and $d \neq 0$.

2. Express formally what it means for a function $f : B^2 \to B$ to be θ-consistent for an equivalence relation θ on B.

   Hint: For example, the function sum defined in question 1 being $\sim$-consistent captures the fact that when adding two rational numbers x and y, we can add two **arbitrary** fractional representations of x and y from the set A (by applying sum), and the rational number obtained does not depend on which representation of x and y was used.

Solution: f is θ-consistent if and only if

$$(b_1 \;\theta\; b_1' \text{ and } b_2 \;\theta\; b_2') \implies f(b_1,b_2) \;\theta\; f(b_1',b_2')$$

is true for all $b_1, b_2, b_1', b_2' \in B$. Alternatively (and equivalently) we could say that f is θ-consistent if and only if

$$([b_1]_\theta = [b_1']_\theta \text{ and } [b_2]_\theta = [b_2']_\theta) \implies [f(b_1,b_2)]_\theta = [f(b_1',b_2')]_\theta$$

is true for all $b_1, b_2, b_1', b_2' \in B$.

3. Prove that the function $\mathsf{sum} : A^2 \to A$ defined in question 1 is $\sim$-consistent.

Solution: Let $(a,b),(a',b'),(c,d),(c',d') \in A$ be arbitrary. We have

$$\begin{aligned}
&(a,b) \sim (a',b') \text{ and } (c,d) \sim (c',d') \\
&\quad\dot{\iff} ab' = ba' \text{ and } cd' = dc' && \text{(def. } \sim\text{)} \\
&\quad\dot{\implies} ab' \cdot dd' + cd' \cdot bb' = ba' \cdot dd' + dc' \cdot bb' \\
&\quad\dot{\iff} ad \cdot b'd' + bc \cdot b'd' = bd \cdot a'd' + bd \cdot b'c' && \text{(comm.)} \\
&\quad\dot{\iff} (ad+bc) \cdot b'd' = bd \cdot (a'd' + b'c') && \text{(distr.)} \\
&\quad\dot{\iff} (ad+bc, bd) \sim (a'd' + b'c', b'd') && \text{(def. } \sim\text{)} \\
&\quad\dot{\iff} \mathsf{sum}((a,b),(c,d)) \sim \mathsf{sum}((a',b'),(c',d')). && \text{(def. sum)}
\end{aligned}$$

Hence, sum is $\sim$-consistent.

Exercise 5.6 Antisymmetry (exam 2021) ($\star$)

This exercise is taken from the end of semester exam of 2021.
Let ρ and σ be relations on some non-empty set A. Prove or disprove:
If ρ is antisymmetric and σ is antisymmetric, then $\rho \circ \sigma$ is antisymmetric.

Solution: Let $A = \mathbb{N}, \rho$ be $\leq$ and σ be $\geq$. $\rho \circ \sigma = \mathbb{N} \times \mathbb{N}$ since for any $a, b \in \mathbb{N}$, there exists k, for instance $\max(a, b)$, such that $a \leq k \wedge k \geq b$. Finally, it is clear that $\mathbb{N} \times \mathbb{N}$ is not an antisymmetric relation since $(1, 2) \in \mathbb{N} \times \mathbb{N}$, $(2, 1) \in \mathbb{N} \times \mathbb{N}$ and $1 \neq 2$.
Tip for the exam: when you have a statement to prove or disprove, do not try to make a proof straight away. Try with a small or understandable example. If the statement does not hold for your example, then you already have a counterexample and the question is over. If the statement holds for your example, you can not deduce anything yet, but you probably understand a bit better the mechanics which is very useful when writing a proof or finding another counterexample.

Due by 23/10/2025, 23:59.
Exercise 5.4 will be graded.